

Design of a 1200 W LLC Resonant DC-DC Converter (400 V to 48 V)

ELECTRICAL SPECIFICATIONS

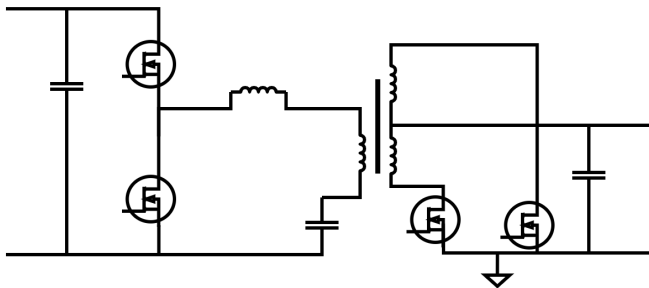
Input Voltage	360V – 400V (nom. 380V)
Output Voltage	42V – 54V (nom. 48V)
Output Power	1200W
Switching Freq.	95 kHz to 150 kHz
Output Ripple	< 0.25V pk-pk
Insulation Resistance	> 100MΩ
HV Side Discharge	Below 60V @ 5s
Efficiency	Up to 91% at full load
Ambient Temperature	-20/60°C

HARDWARE AND SOFTWARE TOOLS

Schematic and Layout	kiCad (PCB-Maker : JLCPCB)
Simulation	LTspice / Ngspice / PySpice
Calculation and reporting	Python / Jupyterlab / VsCode
Firmware Development	STM32CubeIDE and Arduino IDE
Measurement Equipment	TDS3034b, TCP202, Yokogawa 700924

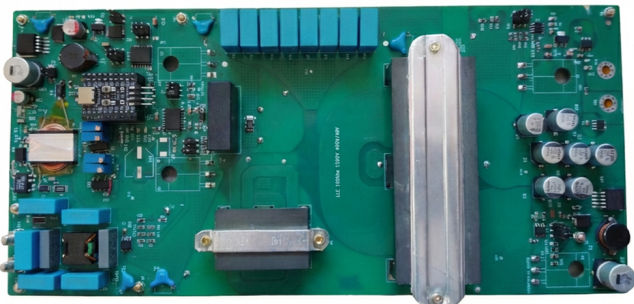
SCHEMATIC / TOPOLOGY

Primary Topology	Half Bridge
Secondary Topology	Synchronous rectifier with center tap
Magnetic cores	TDK Planar ELP
Communication	Isolated SPI 9Mbps
Power supplies	FLyBack 400V/12V, buck 12V/3.3V, buck 48V/12V, LDO 12v/3.3v



ASSEMBLED DC-DC CONVERTER

Dimensions	300 mm x 140 mm x 47 mm
Weight	1.2 kg
Cooling	Forced Air Cooled Heatsinks
Winding assembly	Planar Transformer and Inductor
MOSFET thermal interface	Horizontal Tab-Up, Thermal pad 0.3 mm



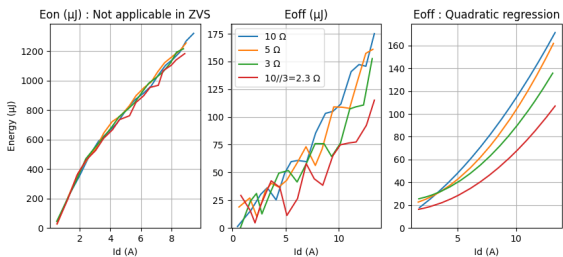
KEY COMPONENTS

Controller IC	STM32F103C8T6 (MINI)
Primary MOSFET (TO-247-3)	IPW60R037P7 600V 76A 37mΩ
Second MOSFET (TO-247-3)	IRFP4568PBF 150 V 171A 4.8mΩ
Primary Driver	UCC21520Q (Source 4A Sink 6A)
Transformer core	ELP102/20/38 with I102/7/38 N97
Resonant Inductor core	ELP43/10/28 N87
PCB	FR4, 4 layers, 1.6 mm, 35μm/70μm
FlyBack controller and core	TNY264, ELP 22/6/16 with I 22/2.5/16

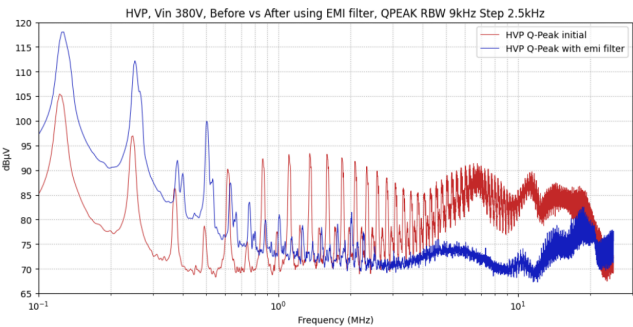
TESTS AND VALIDATIONS

Efficiency	100W up to 1 kW at 380 V
MOSFET characterization	Double pulse test
Magnetic core characterization	Saturation curve
EMI test	Conducted emission 0.15 MHz to 30 MHz
Flyback test	Validation of the design 400V / 12V

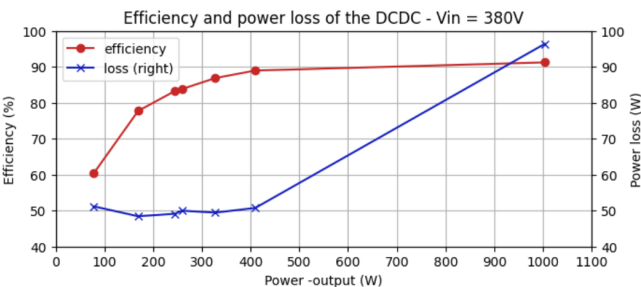
Double Pulse Test



EMI conducted emissions with and without filter



Efficiency Test



PROJECT WEBSITE

For more information, visit the DC-DC website at www.bit.ly/43nwOsP or scan the QR code (More than 300 pages of justification and test reports).

